

Kshitij Singh

AI Student · Backend Engineer

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PROFILE

First-year AI student passionate about building intelligent backends and deploying real-world AI systems. Focused on the intersection of clean software architecture and powerful language models — from autonomous agents to conversational AI. Comfortable working across the full backend stack with Python, FastAPI, and cloud/local LLM integrations.

PROJECTS

Jarvis – Autonomous AI System Agent

Python · Mistral AI · FastAPI · Automation · AI Agent

- Designed an LLM-powered agent capable of interpreting user commands, planning multi-step actions, and executing them through integrated system tools and APIs.
- Built a modular tool architecture enabling the agent to call OS commands, web APIs, and automation scripts dynamically.
- Integrated Mistral AI as the reasoning engine with a custom schema and response manager for structured outputs.

WhatsApp AI Chatbot using Twilio

Python · Twilio API · Webhooks · FastAPI · REST APIs

- Deployed a production chatbot on WhatsApp by integrating Twilio webhooks with a FastAPI backend for real-time message handling.
- Implemented AI-generated response pipeline processing inbound messages through language model inference and returning structured replies.
- Demonstrated scalable conversational AI deployment on a real-world messaging platform with 24/7 uptime.

API Integration & Backend Services

FastAPI · Flask · REST APIs · Python · JSON

- Built a backend platform connecting multiple external APIs and AI services into a unified, scalable workflow.
- Designed request routing and response processing layers enabling clean data exchange between heterogeneous services.

TECHNICAL SKILLS

Languages

Python · C++ · JavaScript · SQL

AI & Machine Learning

Machine Learning · NLP · LLMs · AI Agent Systems

Prompt Engineering · Model Training & Evaluation

AI Frameworks & Tools

Mistral AI · Scikit-learn · NumPy · Pandas · Jupyter

Backend & APIs

FastAPI · Flask · REST APIs · Webhooks

Authentication · JSON APIs

Tools & Platforms

Git & GitHub · Netlify · Linux / Terminal

Competitive Programming

LeetCode **100+ problems solved** · C++ · DSA

EDUCATION

Bachelor of Technology – Artificial Intelligence

2025 – Present

Currently pursuing first year. Focus on AI systems, backend engineering, and applied ML.

CURRENTLY LEARNING

AI Agents & Autonomous Systems · Advanced Machine Learning · Distributed Systems